`blend_fs`: dynamic reaction-limit closure for two-stream reactors

System: `input_fuller_BC_azo_coupling_kinetics` — $\varepsilon = 1$ W/kg, $t = 0.0576$ s ($\approx$50% A conversion)

1 Overview

`blend_fs` constructs a piecewise-linear concentration profile $C_i(f)$ over mixture fraction $f \in [0, 1]$ that tracks the *reaction limit* dynamically as the ODE composition evolves. Rather than committing to a single stoichiometric subset, it blends $K \geq 1$ user-specified "one-short" subset limits, weighting them by the concentrations of their distinguishing products in the current ODE state. The blend determines a kink position f_{sb} and a peak height v_i^{fs} for each species. A scalar $\lambda_i \in [0, 1]$ then interpolates between the no-reaction line and the limit profile so that the Beta-weighted mean matches the ODE value.

The K subsets are prescribed by the user via the `blend_subsets` key in the JSON configuration file (e.g. `"blend_subsets": [20, 31]`). Each entry is an integer index into the enumeration of all reaction subsets; any number of subsets $K \geq 1$ may be listed.

2 Background: mixture fraction and Beta distribution

Mixture fraction f is the stream-1 fraction. Feed concentrations:

$$Y_1: \ A = 15 \text{ mol/L}, \quad \text{others} = 0 \qquad Y_2: \ B = C = 1.2 \text{ mol/L}, \quad \text{others} = 0$$

with mean $\bar{f} = 0.0625$. At time t the PDF is Beta(α, β):

$$s = \frac{\bar{f}(1-\bar{f})}{\sigma^2(t)} - 1, \qquad \alpha = \bar{f}\, s, \qquad \beta = (1-\bar{f})\, s \tag{1}$$

At the example point: $\sigma^2 = 0.01622$, $s = 2.613$, $\alpha = 0.1633$, $\beta = 2.4498$, $\tau_s = 0.02476$ s.

3 Subset stoichiometric limits

For reaction subset $k \in \{1, \ldots, K\}$, the *stoichiometric mixture fraction* f_{sk} is the value of f at which the stream-2 reactant is exactly consumed if only subset k's reactions run. At $f = f_{sk}$ each species i attains a maximum concentration c_{ki}^{fs}.

The *no-reaction line* (conservative mixing, no reaction) is

$$M_i(f) = Y_{2i} + (Y_{1i} - Y_{2i})\, f \tag{2}$$

The user prescribes the K subset indices via `blend_subsets`; for this example $K = 2$ (subsets 20 and 31):

Subset	Reactions	f_{sk}	Distinguishing products	c_k^{fs} (non-zero, mol/L)
20	R1, R3, R5	0.1379	R, T	R = T = 0.5172; Q = 1.0345
31	R1–R5	0.1935	S	S = Q = 0.9677

Table 1: Subset stoichiometric limits ($K = 2$ example). Distinguishing products are those present (non-zero) in subset k's limit but absent in all other subsets' limits.

4 Dynamic blend weights

Each subset k has a set $\mathcal{D}_k$ of distinguishing products. For $k = 1, \ldots, K$ the weight is

$$w_k = \frac{\displaystyle\sum_{i \in \mathcal{D}_k} y_i}{\displaystyle\sum_{k'=1}^{K} \sum_{i \in \mathcal{D}_{k'}} y_i} \tag{3}$$

where y_i is the ODE concentration at the current step. The weights sum to unity and shift dynamically as the composition evolves: subsets whose distinguishing products are more abundant receive higher weight.

For this example ($K = 2$) at the example point:

$$w_{20} = \frac{y_R + y_T}{(y_R + y_T) + y_S} = \frac{0.2290 + 0.0083}{0.2290 + 0.0083 + 0.0187} = 0.9271 \qquad w_{31} = 0.0729$$

5 Blended kink position f_{sb}

The kink is placed using *odds ratios* to stay in the stoichiometric frame:

$$s_k = \frac{f_{sk}}{1 - f_{sk}}, \qquad s_{\text{blend}} = \sum_{k=1}^{K} w_k\, s_k, \qquad f_{sb} = \frac{s_{\text{blend}}}{1 + s_{\text{blend}}} \tag{4}$$

The odds ratio $s_k = f_{sk}/(1 - f_{sk})$ is proportional to the A:B molar consumption ratio in subset k. Averaging in odds-ratio space therefore averages the blended stoichiometry, which is the physically correct quantity to interpolate.

For this example ($K = 2$) at the example point:

$$s_{20} = \frac{0.1379}{0.8621} = 0.1600, \quad s_{31} = \frac{0.1935}{0.8065} = 0.2400$$

$$s_{\text{blend}} = 0.9271 \times 0.1600 + 0.0729 \times 0.2400 = 0.16583 \qquad f_{sb} = \frac{0.16583}{1.16583} = 0.1422$$

6 Control-point limit profile $B_i(f)$

The limit profile is a *tent function* — piecewise linear with a single kink at f_{sb}:

$$B_i(f) = \begin{cases} v_i^0 + (v_i^{fs} - v_i^0)\,\dfrac{f}{f_{sb}} & f \le f_{sb} \\ v_i^{fs} + (v_i^1 - v_i^{fs})\,\dfrac{f - f_{sb}}{1 - f_{sb}} & f > f_{sb} \end{cases} \tag{5}$$

The peak is set by the stoichiometric yield re-scaled to f_{sb}:

$$v_i^{fs} = (1 - f_{sb}) \sum_{k=1}^{K} w_k \frac{c_{ki}^{fs}}{1 - f_{sk}} \tag{6}$$

The factor $c_{ki}^{fs}/(1 - f_{sk})$ is the yield of species i per unit of stream-2 reactant consumed in subset k; $(1 - f_{sb})$ is the stream-2 reactant available at f_{sb}. The formula is exact for any K because each subset's yield is independent of the others. The endpoints v_i^0 and v_i^1 are weighted averages of the subset endpoint values; for unfed product species both are zero.

Example ($K = 2$) — species R and S:

$$v_R^{fs} = 0.8578 \times \left[0.9271 \times \tfrac{0.5172}{0.8621} + 0\right] = 0.4771 \text{ mol/L} \tag{7}$$

$$v_S^{fs} = 0.8578 \times \left[0 + 0.0729 \times \tfrac{0.9677}{0.8065}\right] = 0.0751 \text{ mol/L} \tag{8}$$

7 Species profile $C_i(f)$ and λ_i

The full profile interpolates between no-reaction and limit:

$$C_i(f) = (1 - \lambda_i)\, M_i(f) + \lambda_i\, B_i(f) \tag{9}$$

λ_i is chosen so that the Beta-weighted mean matches the ODE value y_i:

$$\lambda_i = \frac{y_i - \mathbb{E}[M_i]}{\mathbb{E}[B_i] - \mathbb{E}[M_i]}, \qquad \lambda_i \in [0, 1] \tag{10}$$

For unfed product species $M_i(f) = 0$, so $\mathbb{E}[M_i] = 0$ and $\lambda_i = y_i/\mathbb{E}[B_i]$.

Species	y_i (mol/L)	$\mathbb{E}[B_i]$ (mol/L)	λ_i (raw)	λ_i (clamped)
R	0.2290	0.1070	2.14	1.0
S	0.0187	0.01682	1.11	1.0

Table 2: Interpolation parameters at the example point. Both species clamp to $\lambda = 1$: the ODE concentrations exceed the Beta-weighted limit-profile mean, so $C_i(f) = B_i(f)$.

8 Rate integrals

The turbulent reaction rate for reaction j is

$$r_j = \int_0^1 k_j \prod_i C_i(f)^{n_{ij}} \ \mathrm{Beta}(f; \alpha, \beta)\, df \tag{11}$$

Because each $C_i(f)$ is piecewise linear in f, the integrand on each segment $[0, f_{sb}]$ and $[f_{sb}, 1]$ is a polynomial in f. The integral over each segment is evaluated analytically using the regularised incomplete Beta function $I_x(a, b) = B(x; a, b)/B(a, b)$.

Rate constants (L mol^{-1} s^{-1}): $k_1 = 12238$, $k_2 = 1.835$, $k_3 = 921$, $k_4 = 22.25$, $k_5 = 124.5$.